

Date	2/11/2025
Time id	NM2025TMID07789
Project name	Calculating Family Expenses Using Service Now
Maximum mark	2 marks

Project design phase

Solution Architecture

1. Goals

The goal of this project is to build a ServiceNow-based automated system that helps families easily manage and monitor their monthly expenses, savings, and budgets without manual effort.

Below are the key goals:

1. To eliminate manual tracking of income and expenses using automated workflows.
2. To categorize family spending into sections such as groceries, rent, utilities, and leisure.
3. To provide monthly and yearly summaries for better financial planning and analysis.
4. To send alerts when spending exceeds budget limits set by the user.
5. To visualize all financial data through interactive dashboards and reports.
6. To enhance financial awareness, encourage savings, and improve decision-making for families.

2. Key Components

Component	Description
Expense Input Module	Collects income and expense details through forms or data integrations.
Classification Engine	Automatically classifies transactions into categories using predefined ServiceNow rules.

Component	Description
Analytics Dashboard	Provides visual reports, graphs, and spending insights for each .
Budget Management Module	Enables users to set monthly or annual budgets and monitor limits.
User Profile Management	Stores information about family members, income sources, and spending preferences.
Notification System	Sends alerts and reminders about expense thresholds and savings targets.
Data Storage & Security	Ensures all financial data is securely stored in ServiceNow tables with restricted access.

3. Development Phases

Phase	Objective	Key Activities	Deliverables
Phase 1: Requirement Analysis	Understand current challenges and define system requirements.	Stakeholder interviews, process mapping, defining success criteria.	Requirement Specification Document
Phase 2: System Design	Develop architecture and data flow models.	Create system diagrams, define modules, select technologies.	Design Blueprint, Data Flow Diagram
Phase 3: Development	Build system modules and integrate functionalities.	Develop frontend, backend, APIs, and database connections.	Working Prototype
Phase 4: Testing & Validation	Ensure system accuracy and reliability.	Unit testing, integration testing, and performance validation.	Test Report, QA Approval
Phase 5: Deployment	Launch the solution in the production environment.	Configure servers, integrate with ITSM platforms, and monitor.	Deployed Application
Phase 6: Monitoring & Maintenance	Continuously improve performance and address issues.	Collect feedback, monitor KPIs, and implement updates.	Maintenance Logs, Performance Reports

Solution Architecture Design

The solution architecture design for the Calculating Family Expenses using ServiceNow project focuses on automating the entire process of income and expense management through an integrated and workflow-driven platform. The architecture begins with an Expense Input Module that gathers data from users or connected sources such as bank statements and digital receipts. This data is processed by a Classification Engine that categorizes transactions into various types like groceries, rent, utilities, or savings using ServiceNow business rules and flow designer. The Budget Management Module monitors spending patterns and triggers alerts when expenses exceed predefined limits, ensuring effective financial control. All data is securely stored within ServiceNow tables under the Data Storage and Security layer, maintaining confidentiality and access control. The Analytics Dashboard provides a visual overview of monthly and annual spending through charts, summaries, and trend analysis, helping users make informed financial decisions. Together, these components form a seamless and automated architecture that simplifies expense tracking, promotes transparency, and enhances budgeting efficiency **for families.**